WMCS045-05 2026
Lab Assignment #7
Deadline: 2026-10-14 **Group number: _____**

1. Parallelized memory access

Return to the stream benchmark explored already in lab #2, exercise #1. Make sure that it is compiled with OpenMP enabled. Address the following tasks:

a) Determine the bandwidth as a function of the number of threads. Use two different values for the array length: $N = 1024$ (fit in L1D) and $N \geq 2^{22}$ (no fit in LLC). Try to explain your observations.

b) Explore, using a sufficiently large number of threads, the impact of affinity turning. For this purpose, collect results for the following settings of `OMP_PROC_BIND`: `true`, `false`, `master`, `close`, `spread`. Try to explain your observations and briefly reason about possible benefits and/or disadvantages for each of the settings.

2. OpenMP parallelization and scheduling policies

Consider a matrix-vector multiplication routine:

```
1  void mxv(double *restrict y,
2           double *restrict a,
3           double *restrict x,
4           int m, int n)
5  {
6    for (int i = 0; i < m; i++) {
7      y[i] = 0.0;
8      for (int j = 0; j < n; j++)
9        y[i] += a[i*n+j] * x[j];
10   }
11 }
```

Address the following tasks:

a) Parallelize the routine using OpenMP

b) Which schedule would you propose for this parallelization? Explain your answer and briefly list the pros and cons for each of the three OpenMP scheduling policies (namely static, dynamic and guided) for this specific case.

c) Use the Kunpeng 920 nodes with their large number of cores to check your earlier answers.